

Security in IoT Infrastructures - an MQTT Channel approach for intrusion, and Anomaly detection

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Abstract

Low-powered and resource constrained Internet of Things (IoT) devices are often easy targets for attacks. A compromised device planted in a sensitive network can cause major damage. This study employs machine learning (ML) and artificial intelligence for identifying and thwarting attacks. The study aims to secure networks by analyzing data in the network through the message queuing telemetry transport (MQTT) protocol. MQTT is often the preferred protocol for low latency communication in IoT products but often not secured to keep the power consumption low. We collected MQTT traffic data, analyzed temporal and feature-based patterns, and trained ML models to detect anomalous behaviors. The system is designed to identify known devices by using a “signature” data pattern provided by the manufacturers. Any devices not on this list are detected as unknowns, while unfamiliar behavior from known devices are flagged as potential attacks. The trained models achieve 99.32% accuracy in identifying known devices, detecting unknown devices, and recognizing anomalous behavior, outperforming the previously reported 96.29% accuracy, which focused solely on anomaly detection.

Keywords: Artificial Intelligence, DDoS, Internet of Things, MQTT, Machine Learning, Man in the Middle (MitM)

1. Introduction

1.1. Background

The explosive growth of Internet of Things (IoT) devices has revolutionized our lives and work. The ability to connect and exchange data autonomously has enhanced daily life for individuals and businesses while driving significant innovation [1]. In sectors like digital cities [2], healthcare, home automation [3], and IoT businesses, IoT devices are now commonly used. However, this explosive growth has also brought about new attack points, exposing systems to vulnerabilities and making them attractive targets for cyber attacks [4–6]. IoT security breaches can cause disruptions,

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privacy violations, and critical issues, requiring robust measures to prevent and address threats effectively.

IoT systems use various communication and messaging protocols to ensure secure and reliable data exchange. Some of the most common ones include the extensible messaging presence protocol, the constrained application protocol, message queueing telemetry transport (MQTT), and the advanced message queueing protocol [7]. Each serves a unique purpose but MQTT is especially popular in fields like healthcare monitoring, smart agriculture, home automation, and industrial applications because of its lightweight design and efficient machine-to-machine communication [8]. Using the MQTT protocol without strong security protocols can expose IoT networks to cyber risks.

With IoT networks growing rapidly, keeping data secure is more important than ever. Intrusion Detection systems (IDSs) serve as the first line of defense, continuously monitoring network activity and identifying anomalies in real time [9]. By using ML, IDS can recognize patterns and detect attacks more accurately [10]. As more IoT devices connect, new security challenges arise, making advanced AI and ML solution like TinyML, microcontroller based ML and DL increasingly important. These advancements help maintain strong security while keeping computational requirements low and making them ideal for protecting IoT devices. These solutions allow for real time threat detection on IoT devices, reducing computational load [11]. ML's ability to learn from large dataset has proven useful in many areas, especially for securing IoT systems [12]. Using supervised, unsupervised, and reinforcement learning techniques, ML-based systems can achieve high accuracy in distinguishing between normal and abnormal behaviors, making them a powerful tools for IDSs [13].

Despite MQTT's popularity for its lightweight communication, it lacks built in authentication and encryption, which makes it vulnerable to security threats. Attackers can exploit these vulnerabilities to conduct denial-of-service (DoS) attacks, gain unauthorized access, or intercept communications, putting user privacy and confidentiality at risk [14]. It's important to create an effective IDS to protect the MQTT protocol from weakness. While ML techniques are widely used for attack detection, some methods are still face challenges in accurately identifying attacks [15]. One of the key challenges lies in meaning the large data produced by IoT devices which require efficient resource management and large datasets for training ML models [16]. By focusing on both accuracy and resource optimization, feature selection techniques improve the performance of ML algorithms which makes feature engineering an important factor in creating effective and efficient IDSs for IoT networks [17].

1.2. Contributions

The key contributions of this study are summarized as follows:

- We have introduced an intrusion detection system (IDS) for classifying IoT traffic as normal or abnormal behavior using binary classification, with a focus on improving security in MQTT-enabled IoT networks.
- We have collected real-time traffic data from IoT devices (Online Attendance, Sentry, Uprint, and Smart Plug) using Wireshark and analyzed the dataset to develop a robust anomaly detection framework for MQTT traffic.

- The system was designed to identify known devices using a “signature” data pattern provided by manufacturers. Devices not on this list are flagged as unknown, while unfamiliar behavior from known devices is flagged as a potential attack.
- A system was implemented to provide real-time alerts, reducing the risks to MQTT-based IoT systems by utilizing AI and ML for enhanced security and quick responses to incidents.

1.3. Paper Organization

The structure of this paper is as follows. Section 2 discusses related work and presents existing techniques and models for anomaly detection. Section 3 details the system methodology. Section 4 presents the result of the proposed model and includes a comparative study with other works. Section 5 provides the conclusion.

2. Related Works

Limited processing memory and power in many IoT devices creates significant challenges to implement robust security measures. Consequently, new and unseen vulnerabilities making them susceptible to multiple potential threats such as data breaches and unauthorized access. In this section, we aim to provide a comprehensive overview of current advancements in IoT security, with a particular focus on designing a method to secure the MQTT protocol and ensure the safe and reliable communication of IoT systems.

To detect and mitigate abnormal behavior in the MQTT protocol, Ciklabakkal et al. [18] introduced a lightweight anomaly-based IDS for IoT network systems by exploring various ML algorithms such as autoencoders, single-objective generative adversarial active learning (SO GAAL), Random Forest (RF), Isolation Forest (IF), one-class support vector machines (OCSVM), and K-means clustering. Although these techniques effectively detect unknown attacks, their detection accuracy was not sufficient. In [19], researchers used the CatBoost classifier to develop a multiclass classification model for IDS in MQTT network systems. They tested it on both balanced and imbalanced MQTT datasets and compared it to multiple ML models, including RF, Naive Bayes (NB), neural networks (NN), decision trees (DT), and gradient boosting (GB). Their proposed classifier surpassed all models across various ML metrics; however, there were still areas for improvement in the model detection performance.

Hinde et al. [20] proposed another ML-based IDS for the MQTT network, where they generate a new dataset that contains both normal and malicious traffic data. In their study, in total six ML algorithms were investigated, including k-nearest neighbors (KNN), logistic regression (LR), support vector machine (SVM), DT, NB, and RF, based on their performance evaluation metrics (Precision, Recall and F1-score). The results indicated that, for detecting MQTT-specific attacks, flow-based features were more crucial than packet-based features, which are primarily suited for traditional networks. In another study[21], researchers evaluated IDS performance in MQTT IoT networks using only KNN, SVM, and RF classifiers based on accuracy and F1-scores. RF classifier demonstrated the best score among the other tested models.

Vaccari et al. [22] gathered data of both malicious and normal traffic and created a dataset named MQTTset to analyze MQTT anomalies in IoT communications. After testing various ML techniques such as multilayer perceptron (MLP), NB, RF, GB, NN and DT, they concluded that their balanced dataset significantly improved in accuracy and F1-scores. Similarly, Siddharthan et

al. [16] introduced another new dataset for training IDS models named SENMQTT-SET. Their experiments collected data from Raspberry Pi and ESP8266 microcontrollers through a router, based on three main scenarios: normal traffic, attacks on subscribers and attacks on brokers. The study investigated multiple ML models such as DT, SVM, KNN, NB, RF, LR and GB to verify. Additionally, they shared a wide set of performance metrics including F1-score, accuracy, receiver operating characteristic (ROC), Matthews correlation coefficient (MCC), attack detection rate (ADR), false alarm rate (FAR), negative predictive value (NPV), and positive predictive value (PPV). The results demonstrated good training time for DT models. However, in terms of performance metrics other advanced models performed better. The author suggested that further improvements of traffic classifiers could be achieved through DL.

Moreton et al. [23] explored multiclass classification with DL to improve attack detection in MQTT-based IoT networks. The authors used XGBoost, GRU, and LSTM to detect attacks such as DoS, MitM, and intrusions. Their study found that XGBoost achieved the highest accuracy, whereas LSTM and GRU performed better among linear models. In terms of advanced feature selection, an entropy-based method was introduced to improve intrusion detection system (IDS) accuracy [24]. However, the proposed algorithm required further refinement and additional training to enhance classifier performance.

Zeghida et al. [25] proposed an ensemble learning-based IDS model to enhance security in MQTT-driven IoT networks. Their work leveraged multiple ML methods for intrusion detection within the MQTT framework. Comparative evaluations showed that the stacking ensemble learning approach outperformed others, particularly in accuracy, F1-score, and MCC. In [26] a ML based IDS framework was explained to enhance the security in MQTT driven IoT networks. Their approach involved creating a balanced MQTT dataset. Also employing effective feature engineering to optimize detection performance. Upon selecting a model, their system achieved over 96% accuracy, recall, precision, ROC, and F1-score significantly outperforming prior works using the same dataset. Moreover, their framework demonstrated enhanced training and testing times, making it highly efficient for real-time MQTT intrusion detection.

In this study, we propose a solution that analyzes MQTT traffic collected from various IoT devices to easily identify known and unknown devices while comparing the performance of multiple ML algorithms. Additionally, our aim is to develop a real-time infrastructure to check for normal and abnormal behavior if the device is known and create a centralized administrative panel to assist in identifying potential threats.

3. Methodology

3.1. Proposed Model

This model presents a classification method for MQTT network communication, aimed at classifying it as normal or abnormal and identifying known and unknown devices. The goal is to enhance the IDS by using ML models for detecting attacks and classifying devices. In this study, we used five physical devices: a Raspberry Pi 3B+, Sentry, Uprint Kiosk, an Online Attendance System and a Smart Plug, as shown in Fig. 1. Data from other devices was collected from the CICIoT Dataset 2023 for MQTT analysis purposes. The study incorporates various ML models, including IF, DT, RF, KNN, and AdaBoost. This research primarily investigates key features for detecting malicious activities and identifying devices, aiming to improve detection accuracy, reduce overfitting, and distinguish between known and unknown devices. We also optimize hyperparameters to



Figure 1: IoT devices used in the experiment for network behavior analysis and anomaly detection, including both known and unknown devices for evaluation.

fine-tune the ML classifiers for optimal performance. The proposed model is evaluated in terms of performance metrics, such as accuracy, recall, precision, and F1-score. Fig. 5 demonstrates the IDS for MQTT traffic, detecting real-time device activity, identifying both known and unknown devices and notifying the admin panel when potential attacks are detected.

3.2. Environment Tools

We used Jupyter Notebook, a powerful and interactive environment from the Anaconda platform widely recognized in AI research [27], to develop our model. Python was chosen for its popularity in ML, offering effectiveness, stability, ease of evaluation, and scalability. To collect data from the router, we relied on OpenWRT and Tcpdump, a trusted tool for capturing network traffic and the DPKT library to efficiently parse packets and extract key features. For data handling, Pandas and NumPy streamlined processing, making it easy to transform raw network data into meaningful insights.

3.3. Dataset Generation

To establish a well-structured dataset, IoT devices were initially connected to the router, and its normal traffic was captured using Wireshark. The recorded network packets were then labeled to distinguish different traffic patterns. Finally, the labeled data was structured for dataset processing. Fig. 2 illustrates the complete workflow, from data collection to structured dataset generation.

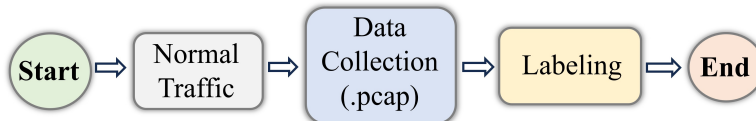


Figure 2: Baseline traffic workflow: Capturing normal traffic (PCAP), labeling data and preparing datasets for ML-based classification in MQTT traffic.

After capturing normal traffic, we execute attacks such as DoS or DDoS and capture the resulting malicious network traffic in PCAP format. The captured traffic is labeled to separate normal

data from attack data. Once labeled, the dataset is structured for further analysis. Fig. 3 represents this process, showing each step from executing the attack to creating the dataset.



Figure 3: Workflow for attack scenario analysis: Executing attacks, capturing malicious traffic (PCAP) labeling data and preparing datasets for ML-based IDS monitoring on MQTT traffic.

3.4. Dataset Processing

The data processing step shown in Fig. 4 begins by capturing both attack and benign traffic using Tcpdump, a powerful command-line tool for network packet analysis. Tcpdump efficiently records raw network traffic which provides a basis for deeper inspection. To extract meaningful insights from these packets, we use the DPKT package.

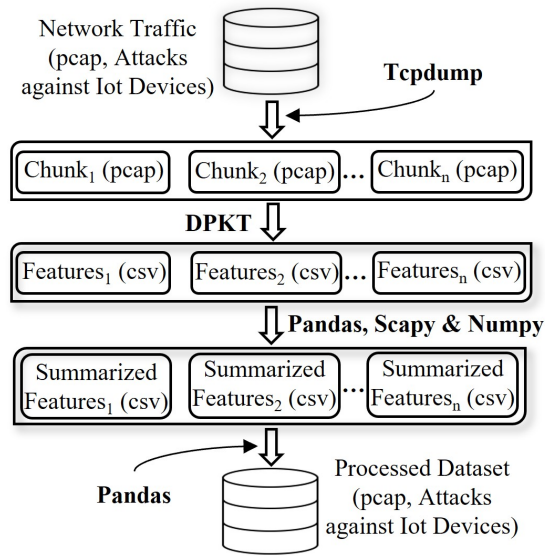


Figure 4: Illustrating the process of chunking total network traffic by device MAC address, converting each chunk into CSV format, and preparing the processed dataset for evaluation.

A parallel process extracts key traffic features with DPKT, saving them in separate comma-separated values (CSV) files, as detailed in Table 1. DPKT enables flexible feature extraction, focusing on key IoT attributes while removing incomplete packets to ensure data quality. The diagram outlines the dataset creation process: data is generated during real attacks, extracted, labeled, processed for accessibility, and evaluated using ML models for classification tasks.

3.5. Machine Learning Evaluation

After the data is preprocessed, it is structured, divided into training (80%) and testing (20%) sets, normalized for consistency, and key features are selected to enhance classification accuracy.

Table 1: 43 MQTT parameters extracted from network traffic, representing the key features utilized for final feature selection process.

No.	Feature	Description
1	tcp.time_delta	TCP Time Stream
2	tcp.flags	TCP Flags
3	tcp.len	TCP Segment Length
4	Protocol Type	IP, UDP, MQTT, IGMP, ICMP, TCP
5	duration	Time to Live (TTL)
6	rate	Packet Rate
7	mqtt.msg	Message
8	mqtt.dupflags	DUP Flag
9	mqtt.len	Msg Len
10	mqtt.hdrflags	Header Flags
11	mqtt.msgid	Message Identifier
12	mqtt.qos	QoS Level
13	mqtt.conack.flags	Acknowledge Flags
14	mqtt.msg	Message
15	mqtt.msgtype	MQTT Message Type
16	mqtt.protoname	Protocol Name
17	mqtt.proto_len	Protocol Length
18	mqtt.conflag.cleansess	Clean Session
19	mqtt.conack.val	Acknowledgement Value
20	mqtt.ver	Protocol Version
21	mqtt.conflags	Connection Flags
22	mqtt.conf.username	Connection Username
23	mqtt.conflag.retain	Retain Flag
24	mqtt.conflag.passwd	Connection Password
25	mqtt.kalive	Keep-Alive
26	mqtt.conflag.willflag	Indicating “Will” Flag
27	mqtt.sub.qos	Quality of the Service
28	mqtt.conflag.retain	Retain the last Message Flag
29	mqtt.conflag.reserved	Reserved Flag
30	mqtt.conflag.qos	QoS Level Flag
31	mqtt.willtopic	Indicates “Will” Topic
32	mqtt.conack.flags.sp	Acknowledgement Session Present Flag
33	mqtt.conack.flags.reserved	Acknowledgement Reserved Flag
34	mqtt.conflags.willmsg	Indicates “Will” Message
35	mqtt.suback.qos	QoS level for Subscription acknowledgement
36	mqtt.willmsg_len	Indicates “Will” Message Length
37	mqtt.willtopic_len	Indicates “Will” Topic Length
38	AVG	The average length of the packet
39	Std	Packet length standard deviation
40	Tot size	Length of the data
41	IAT	Time difference between the previous packet
42	Number	Number of packets
43	Variance	Incoming/outgoing packet length variance

These steps are crucial for improving the performance of the ML models. Next, ML models such as DT, RF, KNN and AdaBoost are used to classify whether a device is known or unknown. These models analyze past traffic patterns and learn to distinguish between different types of behavior. When new traffic is detected, the system first determines if the device is known or unknown. If it is a known device, it undergoes further analysis using the IF model, which checks whether its behavior is normal or suspicious. Finally, the system determines the outcome. If the traffic is normal, the device is classified as safe. However, if the system detects suspicious activity either from an unknown device or abnormal behavior from a known device it flags the traffic as a potential attack. When this happens, an alert is immediately sent to the admin panel, ensuring that security teams can take swift action to investigate and mitigate the threat.

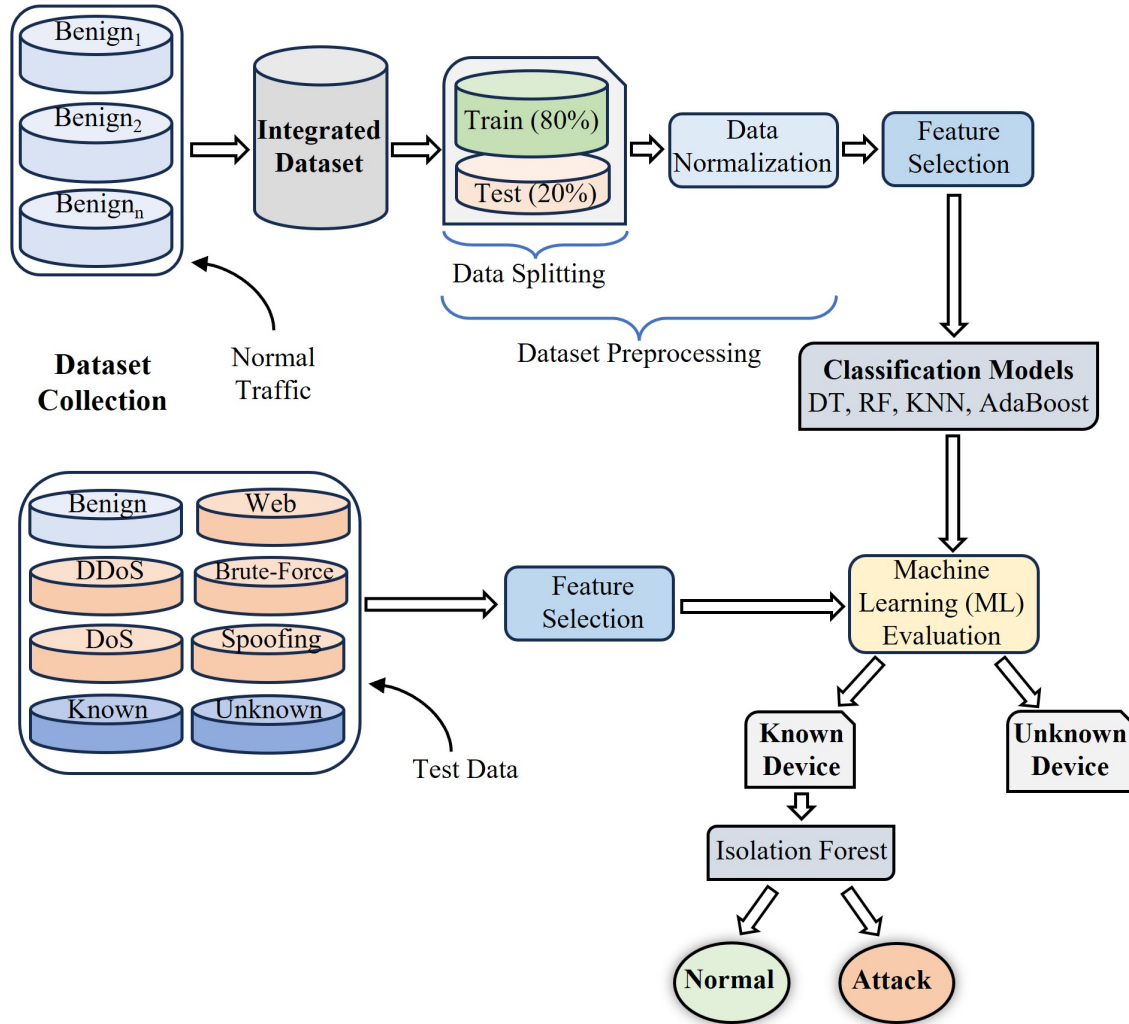


Figure 5: Illustrating the detection of normal and malicious traffic, identification of known and unknown devices, and recognition of potential attacks in an MQTT network.

3.6. Feature Selection and Visualization of TCP Time Stream

Feature selection techniques were used to analyze datasets with statistical methods and ML, which enhance IDS and simplify the model [16]. By identifying the most relevant features and

discarding less significant ones, feature selection improves performance, reduces overfitting, and streamlines the modeling process [28]. In this work, we utilized the feature list from the approach

Table 2: The final set of 10 selected features for device identification and anomaly detection after the feature selection process.

No.	Feature	Description
1	tcp.time_delta	TCP Time Stream
2	tcp.flags	TCP Flags
3	tcp.len	TCP Segment Length
4	mqtt.msg	Message
5	mqtt.dupflags	DUP Flag
6	mqtt.len	Msg Len
7	mqtt.hdrflags	Header Flags
8	mqtt.msgid	Message Identifier
9	mqtt.qos	QoS Level
10	mqtt.conack.flags	Acknowledge Flags

outlined in [26] for evaluation. These features were selected by their method and serve as the most relevant attributes for classification tasks. Table 2 provides an overview of the final selected features along with their descriptions.

Among the top features listed in Table 2, TCP Time Stream stands out for its importance in identifying network traffic patterns. To visualize this behavior, we include the TCP time stream interval graph in Fig. 6. This graph shows the time difference between two consecutive packets, highlighting the distinct patterns of normal traffic generated by each device. Such temporal features are essential for improving the accuracy of both network traffic analysis and device classification and anomaly detection.

4. Result and Discussion

4.1. Performance Measurements

To assess the weaknesses and strengths of the proposed model, the performance is evaluated in terms of some metrics like accuracy, recall, precision, and F1-score. These metrics are defined as follows based on the values of the confusion matrix, like true positive (TP), false positive (FP), true negative (TN), and false negative (FN) [26]. Here, TP represents how often the model accurately classifies positive MQTT network traffic instances as attacks, whereas FP represents how often the model wrongly classifies negative MQTT network traffic instances as attacks. Moreover, TN represents the number of times our model correctly classifies negative MQTT network traffic instances as normal, whereas FN represents the number of times our model incorrectly classifies positive MQTT network traffic instances as normal.

4.1.1. Accuracy

It represents the ratio of correctly predicted samples to the total number of samples. It is calculated as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \quad (1)$$

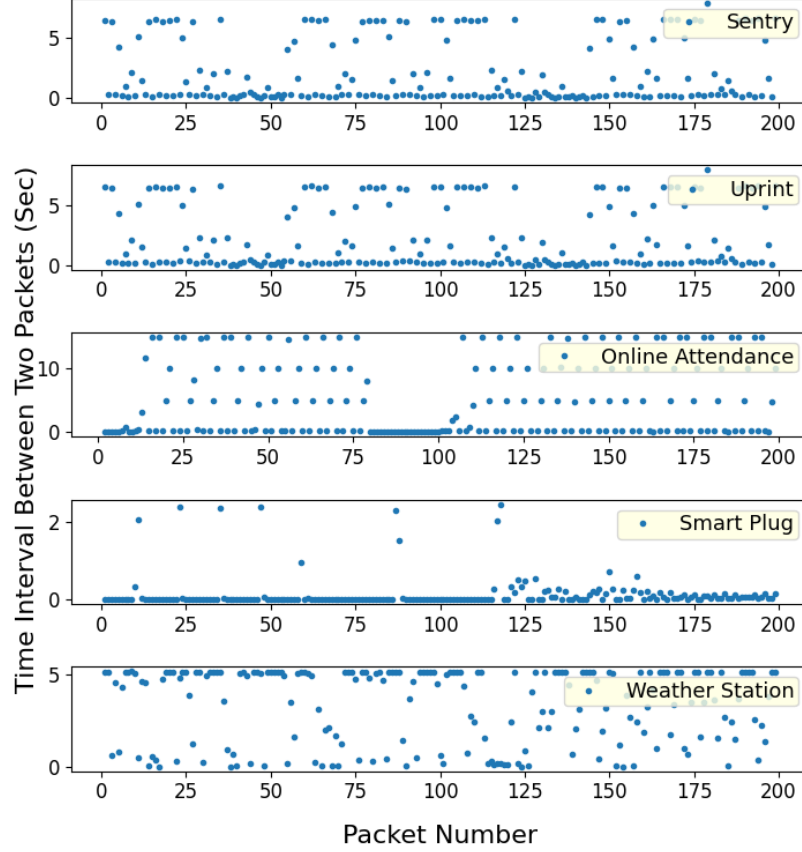


Figure 6: Time interval analysis between consecutive packets for five IoT devices (Sentry, Uprint, Online Attendance, Smart Plug, and Weather Station), showcasing unique temporal patterns characteristic of each device.

4.1.2. Precision

It measures the proportion of true positive predictions out of all instances predicted as positive. Precision is calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

4.1.3. Recall

Recall is a metric that measures the proportion of actual positive instances that are correctly identified by the model. Recall is calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

4.1.4. F1-Score

It is defined as the weighted average of precision and recall, and it is calculated using the mean of the harmonic. F1-Score is calculated as follows:

$$\text{F1-Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Table 3: Showing the optimized hyperparameters used for training the ML model, ensuring improved performance in device identification and anomaly detection.

Algorithm	Best Hyperparameters
DT	{‘criterion’: ‘gini’, ‘max_depth’: None, ‘min_samples_leaf’: 1, ‘min_samples_split’: 2}
KNN	{‘n_neighbors’: 9, ‘p’: 2, ‘weights’: ‘distance’}
RF	{‘max_depth’: None, ‘min_samples_leaf’: 2, ‘min_samples_split’: 5, ‘n_estimators’: 55}
IF	{‘contamination’: 0.1, ‘random_state’: 42}
AdaBoost	{‘base_estimator’: None, ‘learning_rate’: 1.0, ‘n_estimators’: 90}

4.2. Discussion and Analysis

4.2.1. Hyperparameter

Hyperparameter adjustment is essential for maximizing the effectiveness of ML models and enabling fair comparison. Optimizing these parameters, which are set before training, enhances model accuracy [29]. In this study, we experimented with different parameter settings for various ML models, including DT, KNN, RF, IF, and AdaBoost. After multiple experiments, we identified the best-performing hyperparameters, which are summarized in Table 3.

Table 4: Performance comparison of anomaly detection and device identification models using 43, 20, and 10 features.

Features	43			
Methods	DT	KNN	RF	AdaBoost
Accuracy	0.9833	0.9831	0.9833	0.9833
Precision	0.9843	0.9837	0.9843	0.9843
Recall	0.9833	0.9833	0.9843	0.9833
F1-Score	0.9846	0.9847	0.9846	0.9845
Features	20			
Methods	DT	KNN	RF	AdaBoost
Accuracy	0.9873	0.9877	0.9877	0.9877
Precision	0.9873	0.9867	0.9872	0.9873
Recall	0.9873	0.9871	0.9873	0.9877
F1-Score	0.9876	0.9877	0.9877	0.9868
Features	10			
Methods	DT	KNN	RF	AdaBoost
Accuracy	0.9930	0.9929	0.9932	0.9930
Precision	0.9929	0.9928	0.9930	0.9929
Recall	0.9928	0.9930	0.9929	0.9929
F1-Score	0.9929	0.9929	0.9930	0.9930

4.2.2. Performance Analysis

First, we monitored the traffic of devices connected to the router for different periods of time, such as 2 hours, 4 hours, and longer, and created a new dataset. Then, we evaluated a total of

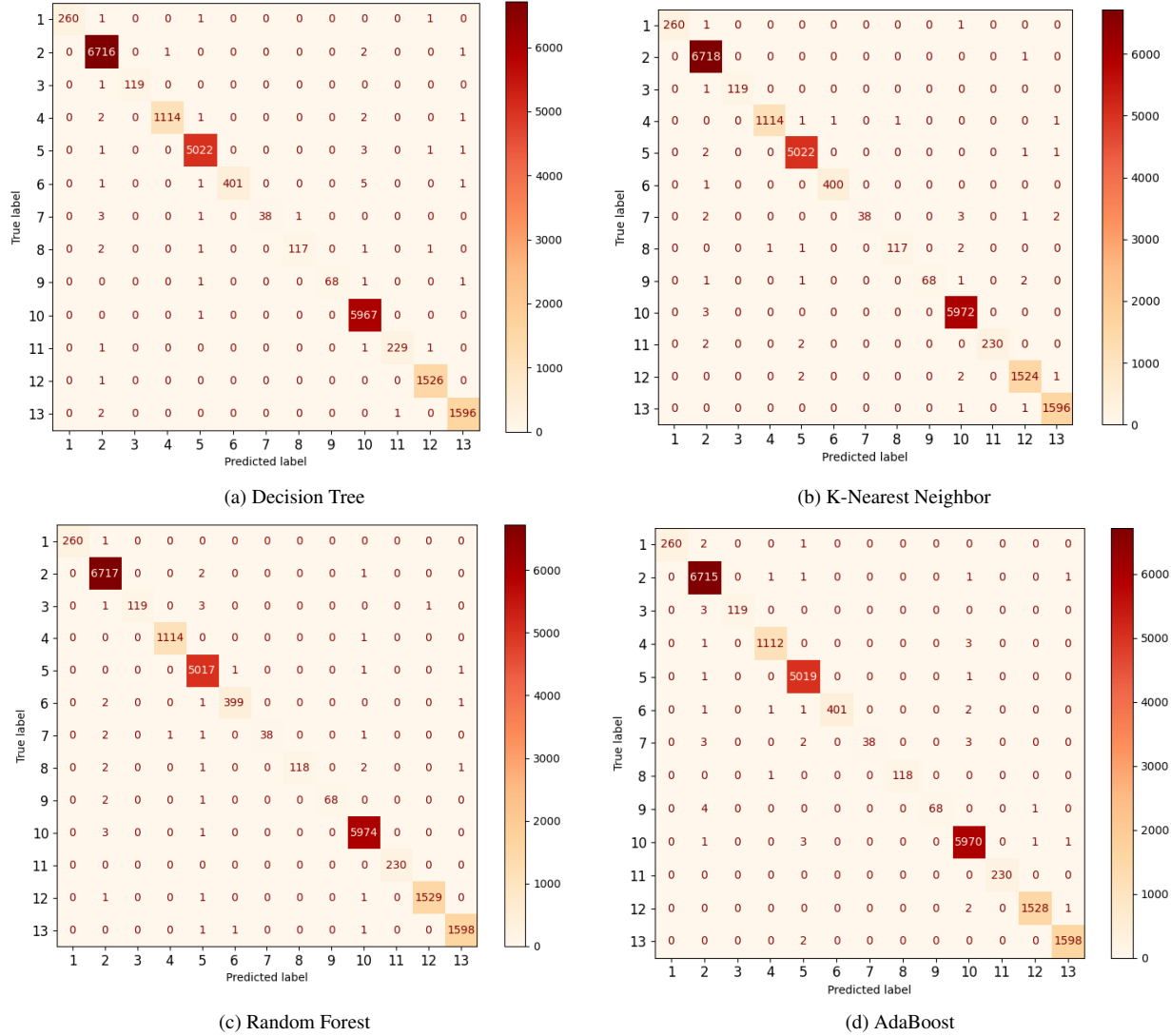
43 features using DT, KNN, RF, and Adaboost. Among these, we obtained the highest accuracy of 0.9833 in DT and RF. For the feature selection process, we followed the method from [26] and selected the top 10 features for evaluation. Using RF, we achieved an accuracy of 0.9932 for identifying known and unknown devices. All results are summarized in Table 4, which includes evaluation metrics such as precision, accuracy, recall, and F1-score. However, the accuracy improved due to hyperparameter tuning and the use of smaller feature sets.

Table 5: Performance comparison between the existing models (33, 31, and 10 features) and the proposed model for anomaly detection and device identification using 10 features.

Paper	[22]			
Year	2020			
Features	33			
Methods	DT	KNN	RF	AdaBoost
Accuracy	0.9159	-	0.9159	-
Precision	-	-	-	-
Recall	-	-	-	-
F1-Score	0.9140	-	0.9140	-
Paper	[25]			
Year	2023			
Features	31			
Methods	DT	KNN	RF	AdaBoost
Accuracy	0.9455	-	0.9538	0.9538
Precision	-	-	-	-
Recall	-	-	-	-
F1-Score	0.9454	-	0.9537	0.9537
Paper	[26]			
Year	2023			
Features	10			
Methods	DT	KNN	RF	AdaBoost
Accuracy	0.9629	0.9627	0.9633	0.9629
Precision	0.9639	0.9636	0.9636	0.9639
Recall	0.9629	0.9626	0.9629	0.9629
F1-Score	0.9629	0.9623	0.9629	0.9629
Paper	Proposed Model			
Year	2025			
Features	10			
Methods	DT	KNN	RF	AdaBoost
Accuracy	0.9930	0.9929	0.9932	0.9930
Precision	0.9929	0.9928	0.9930	0.9929
Recall	0.9928	0.9930	0.9929	0.9929
F1-Score	0.9929	0.9929	0.9930	0.9930

The performance comparison between existing works [22, 25, 26] and the proposed model is presented in Table 5, based on model evaluation metrics (accuracy, precision, recall, and F1-score).

As outlined in Table 5, study [22] utilized 33 features, whereas study [25] used 31 features. This highlights that our approach prioritized the 10 primary features for detecting intrusions, identifying both known and unknown devices, and achieving high accuracy and superior performance across all metrics. Additionally, [26] also used 10 features but did not achieve similar accuracy



1. Sentry 2. Uprint 3. Online Attendance 4. Smart Plug 5. Alexa Echo Dot 6. Amazon Plug 7. Baby Activity Monitor 8. Breast Cancer Detector 9. Amazon Smart Board 10. Amazon Smart TV 11. Surveillance Camera 12. Netatmo Weather Station 13. Coffee Maker

Figure 7: The confusion matrix (a) Decision Tree (DT), (b) K-Nearest Neighbors (KNN), (c) Random Forest (RF) and (d) AdaBoost for identifying known and unknown devices as well as detecting anomalies in MQTT traffic. The list of IoT devices used in the study is provided below the figure.

and precision, demonstrating the effectiveness of our feature selection and model design. Furthermore, a closer look at Table 4 and comparisons with other studies reveal that our model surpasses

the others not only with 10 features, but also with 43 and 20 features. This method optimizes resource utilization, enhances efficiency, accelerates attack response times, and supports the scalability of intrusion detection models. Our scheme outperformed others in areas such as dataset pre-processing, feature selection, data balancing, cross-validation, and hyperparameter optimization, contributing to a robust intrusion detection solution for MQTT IoT networks. While paper

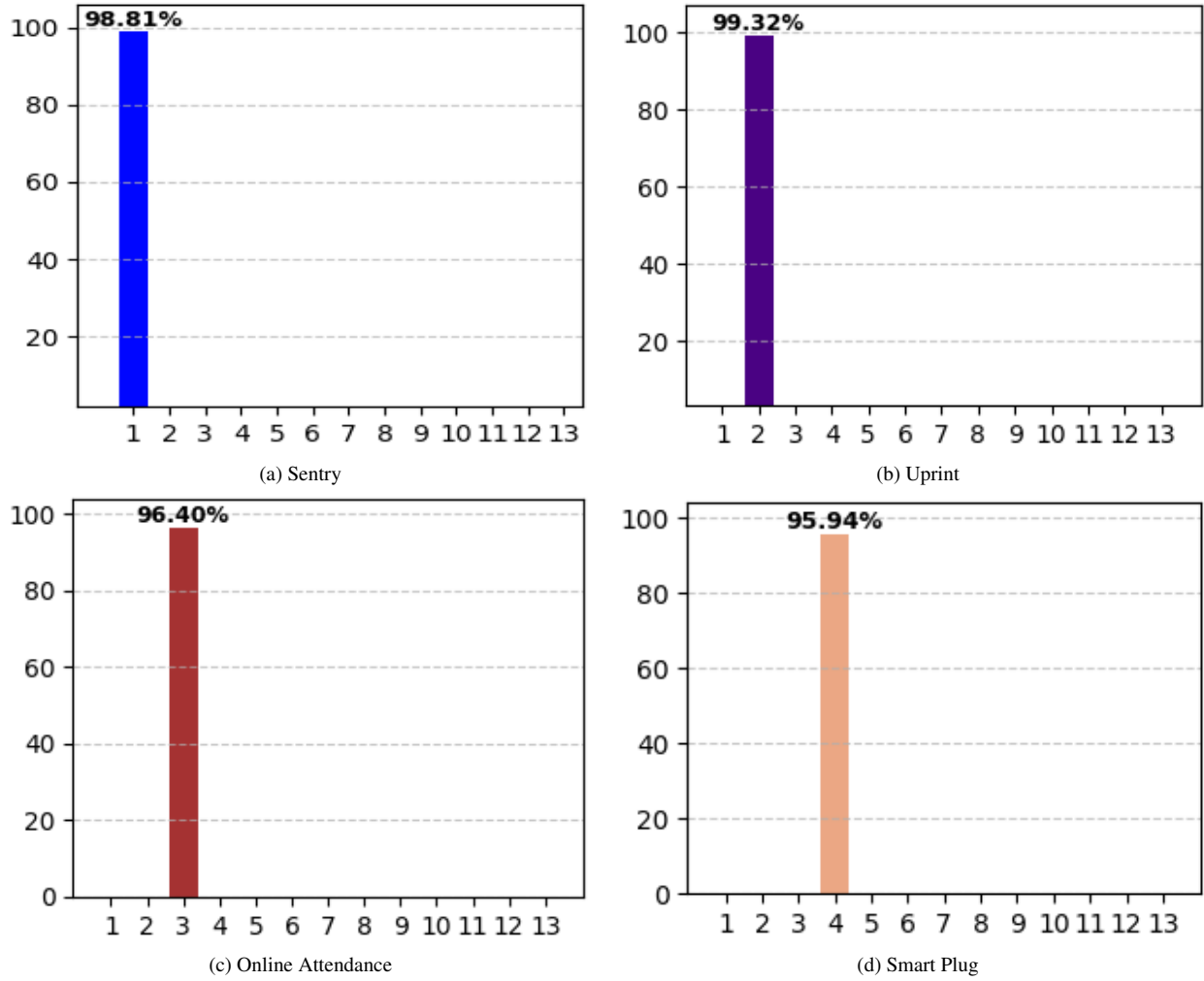
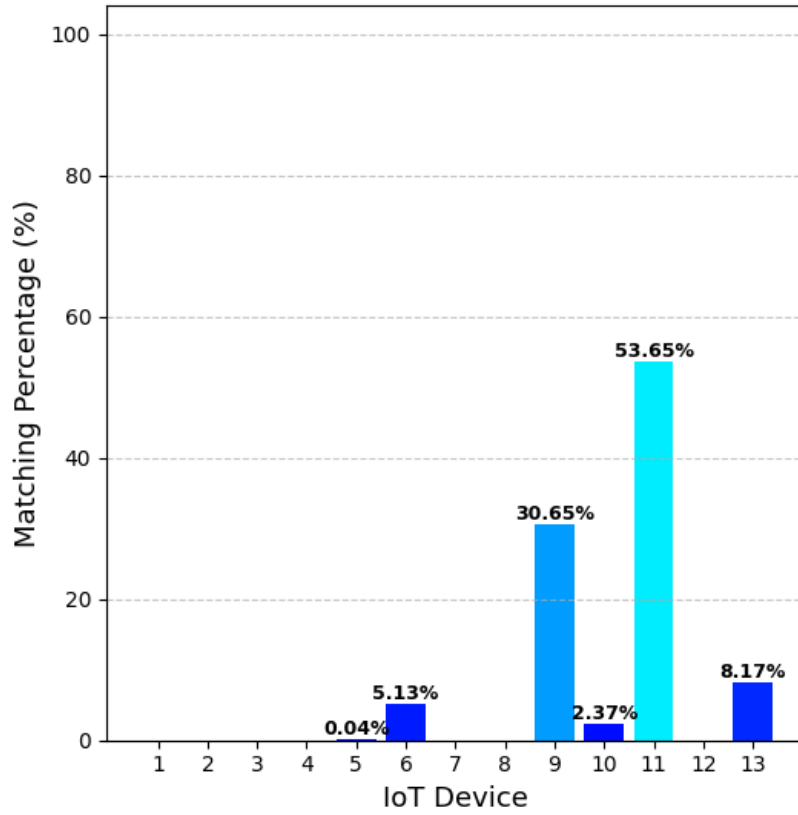


Figure 8: Classification accuracy for (a) Sentry, (b) Uprint, (c) Online Attendance, (d) Smart Plug. Each bar represents the accuracy achieved for identifying known devices, with percentages displayed at the top of each bar. The list of IoT devices used in the study is provided below the figure.

[25] achieved higher accuracy than [22], it did not report hyperparameter for some models, and [22] did not use tuning at all. Neither paper implemented cross-validation. Using large feature sets without clear selection criteria, as seen in [22] and [25], raises concerns about relevance, overfit and

interoperability. Reducing features simplifies the model and removes irrelevant attributes but may lose valuable information. Multiple feature selection techniques can help ensure comprehensive and reliable feature identification.

Fig. 7 depicts the comparison of different ML models DT, KNN, RF and AdaBoost in identifying known and unknown devices and detecting anomalies in MQTT traffic. Each model performed well in distinguishing between devices and identifying abnormal behavior in the network traffic. The matching percentages shown for each IoT device in Fig. 8, such as 99.32% for the ‘Uprint’ device, represent how accurately the device’s MQTT traffic aligns with the model. A 99.32% match, corresponding to an accuracy of 0.9932, indicates that the ‘Uprint’ device’s network behavior closely matches the expected pattern, demonstrating the model’s understanding of the device’s communication. This approach aids in classifying known devices based on their temporal behavior, allowing the ML model to identify anomalies and differentiate between authorized and unauthorized devices. Other devices like ‘Sentry’, ‘Online Attendance’, ‘Smart Plug’ and ‘Alexa Echo Dot’ are similarly evaluated by comparing their MQTT traffic to the expected model.



(a)

1. Sentry 2. Uprint 3. Online Attendance 4. Smart Plug 5. Alexa Echo Dot 6. Amazon Plug 7. Baby Activity Monitor 8. Breast Cancer Detector 9. Amazon Smart Board 10. Amazon Smart TV 11. Surveillance Camera 12. Netatmo Weather Station 13. Coffee Maker

Figure 9: Matching percentage for unknown IoT device identification, illustrating the percentage of similarity achieved for each device during classification.

In this research, we used a pcap file from an ‘Amcrest WiFi camera’ which was not part of the pre-trained combined dataset. Fig. 9 shows a 53.65% match, the closest we could find, with the best match being a known surveillance camera based on MQTT traffic. The second best match was to an Amazon Smart Board, scoring 30.65%. With a 70% threshold, any device below this percentage is considered unknown. Since the best match was only 53.65%, the camera was classified as unknown. Since the highest match was only 53.65%, the Amcrest camera was marked as unknown. A deviation of 46.35% from the closest match also indicates an anomaly, highlighting that the device’s behavior is not consistent with the expected pattern.

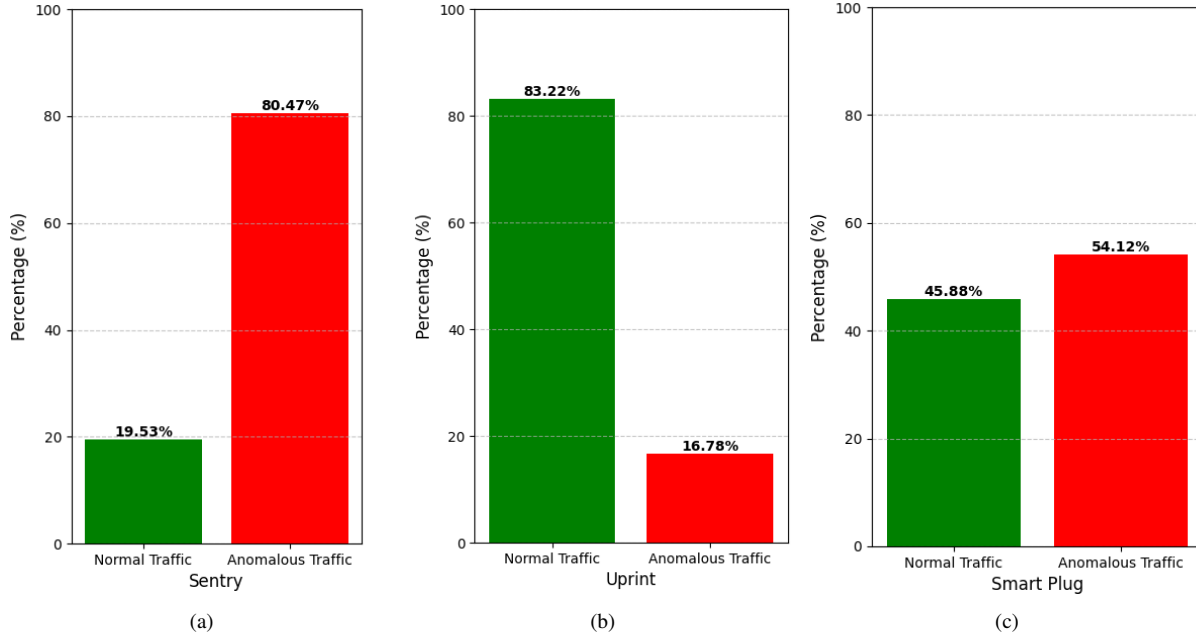


Figure 10: Evaluation of the Isolation Forest model on three devices. (a) **Sentry**: 80.47% of traffic flagged as suspicious, indicating a potential attack. (b) **Uprint**: 16.78% flagged, indicating no significant anomalies (c) **Smart Plug**: 54.12% of traffic flagged, showing a mix of normal and suspicious activity.

Fig. 10 shows the results of applying IF model to analyze the behavior of known devices, specifically Sentry, Uprint and Smart Plug. After identifying these devices, we examined their network traffic individually to determine whether their behavior was normal or potentially malicious. To detect malicious activity, we implemented DDoS attacks of varying duration: 3 minutes for Sentry, 1 minute for Uprint, and 2 minutes for Smart Plug. We captured network traffic in a PCAP file and used it to evaluate abnormal patterns. A 10% contamination rate and a 90% threshold for normal traffic were used in the IF model to detect anomalies. The IF model identified a significant portion 80.47% of Sentry’s traffic as abnormal, indicating a high probability of attack. In contrast, Uprint exhibited mostly normal behavior, with only 16.78% of its traffic classified as abnormal. Smart Plug fell into this category with 54.12% of its traffic identified as suspicious. This approach enhances IoT network security by detecting anomalous communication patterns, improving the IDS’s ability to identify potential threats.

5. Conclusion and Future Work

This research presents a practical IDS for MQTT traffic aimed at enhancing security in IoT networks. Our developed system first identifies devices as either known or unknown. When we recognize the device as known, the IF model is applied to analyze its behavior and detect anomalies. By applying ML, the system effectively classifies IoT traffic, distinguishing typical patterns from potential threats. Through continuous real-time traffic monitoring from various IoT devices, a robust anomaly detection framework is established, contributing to a more secure MQTT-based IoT ecosystem. Five ML algorithms KNN, IF, DT, AdaBoost, and RF were applied to classify MQTT network traffic as normal or abnormal. Among them, RF achieved the highest accuracy of 0.9932, followed by KNN, AdaBoost and DT achieving 0.9930, respectively. The IF model proved particularly effective at identifying anomalies in device behavior. For example, 80.47% of Sentry's traffic was flagged as anomalous, suggesting a high likelihood of an attack, while 54.12% of Smart Plug's traffic was suspicious. Uprint, on the other hand, showed only 16.78% of its traffic as anomalous, indicating mostly normal activity.

In future, a system will be developed that will automatically detect unknown IoT devices when they connect to the network. These devices will be initially classified as unknown and then, if the behavior of the unknown device is not unusual, the system will be trained to detect and register them as known with administrator confirmation. This approach ensures seamless integration while maintaining security and reducing manual effort. Further exploration will include advanced ML and DL techniques to improve efficiency and scalability.

Disclosure statement

The author(s) declare(s) no potential conflict of interest regarding the publication of this paper.

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